

Dalil: an AI arrival guide for international students in Abu Dhabi

Project report for the Hub71+ AI Hackathon supported by OpenAI

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Live prototype: <https://dalil-abudhabi.vercel.app> | Code: <https://github.com/muhammet-tm/hub71ai-team404>

Executive summary

The hackathon challenge asks how to make it easier for people to arrive, settle in, and build a future in Abu Dhabi. This project answers it for one group: international students who hold an admission letter from a university in Abu Dhabi and have not yet settled in the emirate. The guidance these students need exists, but it is spread across the university, federal portals, and Abu Dhabi government services, and the student meets it in pieces.

Dalil (Arabic for "guide") is a deployed web application that turns one photograph of an admission letter into a personal arrival plan. The plan has four phases (Before you fly, First week, First month, and Build a future), it is explained in the student's own language, and every step carries its official source. The student can check a passport against the validity rule on the university's visa form and can ask questions by text or by voice.

The design rests on one rule: the model selects and translates, and a verified file states the facts. Six OpenAI capabilities are used (vision with Structured Outputs, structured generation, streamed grounded answers, moderation, speech-to-text, and text-to-speech), and none of them is allowed to write a fee, a deadline, or a link. When no verified source covers a question, Dalil says so and names the office to ask. The knowledge base holds 18 entries, each traced to an official page and each labeled with the result of an independent fact-check.

The report states its limits plainly. The combined product is unproven, no university office has yet agreed to a pilot, no count of international students in Abu Dhabi was found in any source that could be opened, and the university-specific facts come from a form dated 2018.

1. Problem statement

A literature review of international student challenges lists obtaining accurate information, the admission procedure, and preparing visa documents as the common pre-departure problems, followed by language, financial, and cultural adjustment problems in the host country [1]. In

the United Arab Emirates (UAE), the federal portal states that a student stays either under the sponsorship of a resident parent or under the sponsorship of the accredited university or college, and that the student must submit a certificate from the institution that specifies the duration of study [2]. For a student with no parent in the country, the university is therefore the sponsor and the first point of contact, while identity, residency, and city services sit with federal and Abu Dhabi government bodies.

The only peer-reviewed study of new international students in the UAE that the review found reports that orientation came late. Most participants argued against holding orientation in the second week of the semester and said that information in the first week, or before they arrived, would have helped them prepare [3]. The psychological stakes are documented outside the UAE: a 2024 meta-analysis found a pooled correlation of $r = 0.39$ between acculturative stress and psychological outcomes in international students, and $r = 0.41$ with depression [4].

The problem statement follows from these points. An international student admitted to an Abu Dhabi university has no single plan that is personal to the student's country, university, and arrival date, that ties each step to its official source, and that starts at the admission letter.

The primary persona is Yuki, an 18-year-old from Japan, admitted to Abu Dhabi University (ADU) on a university-sponsored visa. Yuki has intermediate English and arrives in three weeks. Yuki is fictional. The persona is plausible for ADU, where international students made up 59.6% of the student body and represented over 100 nationalities according to an announcement dated 6 June 2023 [5].

2. Context in Abu Dhabi

No count of international students in Abu Dhabi emirate was found in any source that could be opened, and no source publishes the number of students who arrive from abroad each year. This report therefore gives no market size. At the national level, UNESCO data reported by ICEF Monitor give 237,034 international students in the UAE in 2023, up from 54,162 in 2012 [6]. At the institutional level, ADU reported a 59.6% international share in June 2023 [5], and NYU Abu Dhabi lists students from more than 115 countries for 2025-2026 [7].

Abu Dhabi's population reached 4,135,985 in 2024, a rise of 7.5% in one year, and residents aged 25 to 44 make up 54% of the population [8]. The "build a future" part of the challenge has a concrete, sourced milestone for students. The Abu Dhabi Residents Office (ADRO) lists a 10-year Golden Visa for graduates of accredited UAE universities classified A or B, with a

minimum grade point average (GPA) of 3.5 for class A or 3.8 for class B and graduation no more than two years earlier [9]. Dalil shows these thresholds as the final step of the plan from the first day. The argument that a student who settles well is more likely to graduate and remain is reasoning; no retention study was found to quantify it.

Hub71+ AI expects at least one founder of each startup to commit to relocating long-term to Abu Dhabi [10], and 74% of the startups in Hub71's 17th cohort are headquartered outside the UAE [11]. Many founders in the host community are therefore newcomers themselves.

3. Literature review

The review is a rapid scoping review and is not systematic. It retained 16 peer-reviewed sources, and an independent fact-checker confirmed that all 16 exist with the stated authors, year, and venue and corrected the wording of seven claims. Four themes follow the features of the product.

Arrival stressors. Acculturative stress is the best-quantified problem. A systematic review of 29 studies (total N = 7,247 in the meta-analysis) found the pooled correlations quoted in Section 1, and 15 of the 29 studies were conducted in the United States, so the estimate is weighted toward that setting [4]. Evidence from the UAE is thin and qualitative: six focus groups with 40 first-year international students of ten nationalities at one institution [3].

Peer and buddy support. Peer mentoring has the strongest intervention evidence. In Canada, newcomer international students paired with student mentors had higher psychological adaptation and lower acculturative stress than control participants at the end of the program [12]. An Australian field study paired 43 local mentors with 126 international mentees over 18 to 19 weeks and reported effects in cultural adjustment and well-being [13]. In the UAE, a survey of 152 freshmen found that many rely heavily on the support of peers, especially co-nationals [14].

Pre-arrival orientation. Timing is the central finding: students wanted information in the first week or before arrival, and visa delays shift arrival dates [3]. Pre-arrival contact was reported to have a positive effect on engagement and sense of belonging in 31 interviews at a Malaysian university, although that evidence is self-reported and has no comparison group [15].

AI assistants for students and newcomers. The evidence is early-stage. The admissions help desk of Politecnico di Torino built chatbots after receiving more than 20,000 requests from

more than 100 countries since January 2024 [16]. A human-in-the-loop large language model system at one university reported a 30.8% reduction in resolution time and user satisfaction rising from 3.6 to 4.9 out of 5 [17]. RefuGPT combined a language model with retrieval over official and community information for Ukrainian refugees in Switzerland [18]. These studies point to the same architecture: answers grounded in official sources, with a path to a human.

Research gap. No study was found that evaluates an end-to-end AI arrival companion for international students or vision-based document checking for student onboarding, and no peer-reviewed quantitative study measures administrative burden or time to settle for international students in the UAE. Dalil combines components that each have support in the literature. The claim is limited: each component is evidence-informed, and the combined product is unproven.

4. Benchmarking

TAMM is the strongest existing service and sets the standard that Dalil must respect. Its App Store listing describes Abu Dhabi's official government services app with more than 1,000 services, an AI assistant that gives personal guidance, and English plus 59 further languages [19], and TAMM 3.0 added voice interaction in Arabic and English [20]. Artificial intelligence, many languages, voice, and document scanning are therefore present in TAMM already, and none of them distinguishes Dalil. Table 1 compares the closest services.

Table 1. Comparison of the closest existing services, based on the pages opened on 2 October 2026.

Service	Primary user	Personal guidance	Student-specific	Gap observed on the pages opened
TAMM [19]	Abu Dhabi citizens, residents, and visitors	Yes	Not specific	No pre-arrival student journey seen
ICP [21]	Visa and identity applicants	Not stated	Not specific	Transactional; no guidance path seen
ADRO [22]	New residents	No	Golden Visa check only	Static guides
ADUConnect [23]	ADU students	Yes	Yes	Campus life only
Goin' [24]	Universities	Not stated	Yes	Social connection only
NaviSmart AI [25]	Immigrants and students	Yes	Yes	Generic; no UAE or university layer found
Dalil	International students in Abu Dhabi	Yes	Yes	New and unproven

The team's reading of Table 1 is that no service opened joins the stretch between the admission letter and the first settled weeks for one specific student. This is an inference from page text. The report claims only that the team found no such journey on the pages it opened; it does not claim that TAMM cannot provide one. Dalil is positioned as the on-ramp to TAMM for international students: it covers the stretch before the student holds an Emirates ID and then hands the student to TAMM, the Federal Authority for Identity, Citizenship, Customs and Port Security (ICP), and UAE Pass.

5. The solution

The student journey has four screens. Figure 1 shows the journey on the deployed prototype.

The figure displays four screenshots of the DALIL prototype interface, arranged in a 2x2 grid. Each screen features a dark blue sidebar with navigation options: 'Start', 'My plan', 'Ask', and 'Sources'. The top-left screen, titled 'From admission letter to settled in Abu Dhabi', shows a form for 'Check your details' with fields for name, nationality, university, arrival date, and language. It includes a 'Create my arrival plan' button and a sample 'LETTER OF ADMISSION' from Abu Dhabi University. The top-right screen shows a 'JPN' to 'AUH' flight path and a 'Before you fly' section with tasks like 'Know who sponsors your visa' and 'Get a study-duration certificate'. The bottom-left screen, 'Check your passport', includes a 'Check your passport against this rule' section and a 'Passport check' step. The bottom-right screen, 'Ask Dalil', shows a 'Do I need an Emirates ID, and where do I apply?' question with a 'Listen (30 seconds)' button. A progress indicator at the bottom of each screen shows the current step out of 18.

Figure 1. The deployed prototype. Top: the confirmation form filled from the sample letter, and the plan as a boarding pass with the assistant docked beside it. Bottom: the passport check inside its step, and a refusal on the Ask screen.

Start. The student ticks a consent box and photographs the admission letter, or chooses the synthetic sample letter. A vision model reads the letter and fills a confirmation form. Fields the model was less sure about are outlined, and the student edits and confirms every field. The typed form is the guaranteed path, so the plan never depends on a live vision call.

Plan. The plan opens with a boarding pass that names the student, the country of origin, and the arrival date. All 18 verified steps appear at once in English as tickets, grouped in four phases. Each ticket carries a rubber stamp that shows the fact-check status of its entry, and

ticking a step stamps it done. The model then adds a title and one personal sentence per step in the student's language. Each card shows the official fact, the fee where one is confirmed, a caveat where the source is weak, the source with its verification status, and the body that handles the step. Dates are a suggested order unless the card says "Sourced deadline". Inside the documents step, the student can photograph a passport; the model reads the expiry date, and the six-month rule from the ADU visa form is applied in code.

Ask. The student types or speaks a question. The answer streams in the student's language, each claim ends with the marker of the entry it comes from, and the official fact card of every cited entry appears below the answer. A question with no verified source, such as how to open a bank account, receives a refusal and the name of the office to ask.

Sources. The last screen lists the 18 entries with their status, the models in use, what Dalil does not cover, and a button that deletes the student's data from the device.

6. System design and architecture

Dalil is one Next.js application with five stateless server routes, deployed on Vercel. On a desktop screen it fills the window, with a navigation rail on the left and the assistant docked beside the plan; on a phone it collapses to one column. It has no database. The profile, the plan, the checklist, and the chat stay in the browser, and document images are never stored. Figure 2 shows the components.

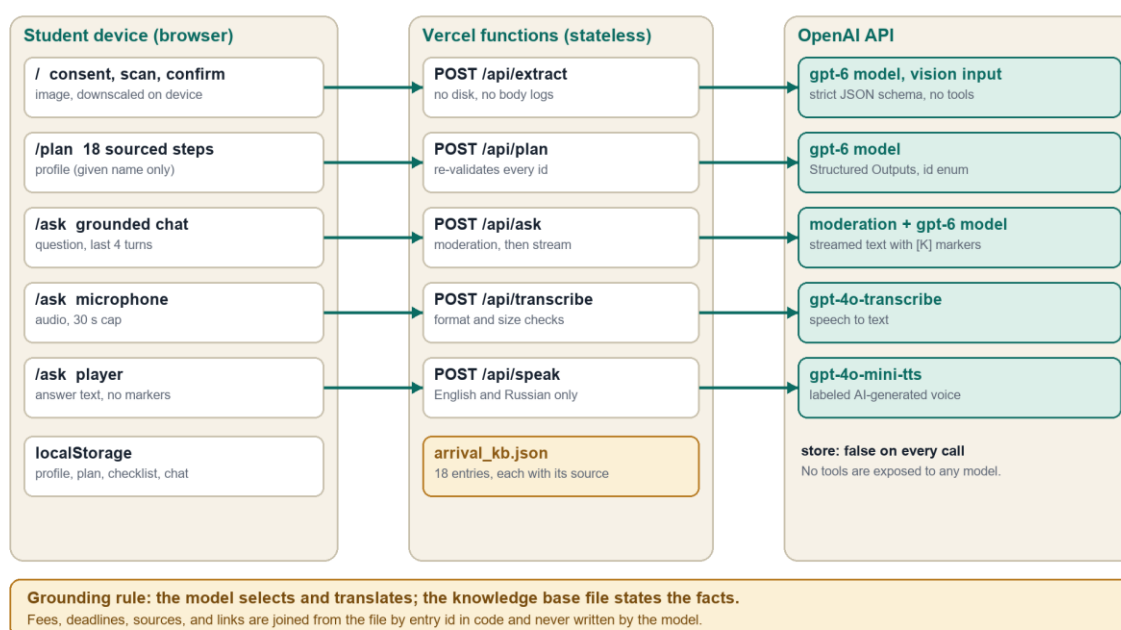


Figure 2. System architecture. The knowledge base file feeds both the interface and the prompts.

The grounding rule is the core of the design. The model returns only an entry identifier, a localized title, and one personal sentence. Facts, fees, deadlines, source names, and links are joined from the knowledge base file by identifier in code. An invented step is blocked twice: the output schema restricts the identifier to the list generated from the file, and the server drops any identifier that is not in the file. Selection is shown as a marker and never as a removal, so a wrong omission by the model cannot hide a required step.

Table 2. OpenAI capabilities used in the prototype.

Feature	Model and API	Guardrail
Read the admission letter and the passport	gpt-6-luna, Responses API, image input, strict JSON schema, one retry on gpt-6-astra	Text in the image is treated as data; no tools; no passport number or date of birth in the schema; human confirmation
Personalize the plan	gpt-6-luna, Structured Outputs	Identifier restricted to the knowledge base; server re-validation; no numbers or links from the model
Answer questions	gpt-6-luna, streaming	Knowledge base only; source marker on each claim; refusal when no source exists
Screen questions	omni-moderation-latest	Flagged input is refused before the model call
Voice input	gpt-4o-transcribe	The transcript is shown for editing before it is sent
Voice output	gpt-4o-mini-tts	Text first; the player is labeled "AI-generated voice"

The text and vision model is set by an environment variable. The tested prototype runs on gpt-6-luna, the efficient model in the current OpenAI family, and the same code runs on gpt-6.1-sol without a change.

The knowledge base holds 18 entries built from official pages of the UAE Government portal, the Abu Dhabi Residents Office, the Department of Health Abu Dhabi, Zayed International Airport, and the ADU visa application form. Each entry records its source, its fact-check status (confirmed, partially confirmed, or dated 2018), and a caveat where the evidence is weak. The plan call is hedged: if the first request has not answered in 9 seconds, a second request starts and the faster one is used. Every live call also has a labeled fallback. If the plan call fails, the screen keeps the English entries under the label "Standard plan". If the chat call fails, the screen shows keyword matches from the file under the label "Offline answer from verified sources". Saved examples of the synthetic sample load only when a live call fails, and they are labeled "Saved example".

7. Privacy, security, and responsible AI

The team is not incorporated, so the design follows the stricter reading of the federal Personal Data Protection Law [26] and the ADGM Data Protection Regulations 2021. Table 3 lists the controls that are built in the prototype.

Table 3. Controls implemented in the prototype.

Control	Implementation
Consent before upload	A checkbox on the start screen; the extract route rejects a request without it
Transfer notice	The consent text names the transfer to a service outside the UAE
Withdrawal	"Delete my data from this device" on the Sources screen
Data minimization	No passport number and no date of birth are extracted; only the given name reaches the model in the plan call
No server storage	Images are held in memory only; every Responses API call sets store to false; no request body is logged
Prompt injection defense	Document text is treated as data; strict schema; no tools; an embedded-instruction flag
Misinformation defense	Facts from the file; source chips; labels on AI-written text; a refusal rule
Input limits	Image type allowlist, 4 MB cap, 500-character question cap

The product and the pitch say that the Dalil server stores no documents. They do not say that nothing is retained anywhere, because OpenAI keeps abuse-monitoring logs for up to 30 days [27], and they do not say that data stays in the UAE, because the prototype does not use OpenAI's UAE data residency option [28]. The threat model follows the OWASP Top 10 for Large Language Model Applications [29], in particular prompt injection, misinformation, and unbounded consumption.

8. Testing and results

The prototype was tested in three ways on 2 October 2026, first locally and then against the production deployment. A script of eleven probes called every server route with the real OpenAI key, in the same way the browser does. A scripted browser session walked through the whole journey on the synthetic sample documents at desktop width (1,440 px) and at phone width (400 px). A third script used a simulated microphone to speak an English question into the page. Table 4 lists the probe results on production. The figures are single runs from one laptop on the venue network, so they show that the features work and do not measure performance.

Table 4. Results of the live probes on the production deployment (gpt-6-luna, called from Abu Dhabi).

Probe	Result	Time
Read the sample admission letter (name, nationality, university, start date, duration)	Pass: all fields correct	2.9 s
Read the sample passport without returning the passport number or the date of birth	Pass: expiry correct, neither value returned	2.6 s
Read a letter that contains a hidden instruction to change the nationality	Pass: instruction flagged and ignored	2.8 s
Personalize the plan in English	Pass: 18 of 18 steps marked, no invented step, no warning	4.8 s
Answer "Do I need an Emirates ID, and where do I apply?"	Pass: answer cites the Emirates ID entry	1.8 s
Answer "How do I open a bank account?"	Pass: refused, with a handoff to the university office	1.3 s
Question that orders the model to ignore its rules and state a fee and a link	Pass: refused, no fee and no link	1.3 s
The Emirates ID question asked in Spanish	Pass: answered in Spanish, cites the Emirates ID entry	1.4 s
Text-to-speech in English	Pass: audio returned	1.6 s
Speech-to-text on that audio	Pass: transcript matches the question	1.4 s

The browser sessions confirmed the same behavior in the interface at both widths. The sample letter filled the confirmation form, the plan appeared at once from the verified file and was then personalized by the model, the sample passport raised the red flag with its source and its age, a ticked step survived a page reload, the refusal notice appeared for the bank account question, and the delete button removed the profile from the device. The voice session recorded the spoken question, showed the transcript for editing, returned a cited answer, and produced the spoken answer with its "AI-generated voice" label. One desktop session logged a single network protocol error on a resource request over the venue network; every step of that session still passed. No page overflowed horizontally at seven widths from 375 px to 1,920 px.

Testing changed three design decisions. First, about one plan request in seven took 20 to 50 seconds against 5 to 8 seconds for the rest, so the plan route now starts a second request if the first has not answered in 9 seconds and uses whichever finishes first; in eight consecutive runs after the change the slowest took 18.2 seconds. Second, gpt-6.1-sol took 30.8 seconds to build the plan in one test against about 6 seconds for gpt-6-luna with comparable output, so the deployment uses gpt-6-luna. Third, gpt-transcribe took 32 seconds on a five-second clip against 1.6 seconds for gpt-4o-transcribe, so voice input uses gpt-4o-transcribe.

No test involved a real student, a real document, or a second language reviewer. The team read the English output. The Spanish answer was checked by one probe, and the Portuguese, Chinese, and Japanese output was not reviewed by a speaker of those languages.

9. Limitations

The analysis contains no primary research. It draws on no student interviews or surveys, no enquiry data from a university office, and no statement of demand from any international office at the time of writing. The persona is fictional. The literature review was completed in one morning with one search tool, and most sources were checked at abstract level.

The knowledge base is thin where it matters most to the persona. ADU's live visa pages could not be opened, so every ADU fact comes from one form dated 11 February 2018, and no current ADU fee is shown. No official fee was confirmed for the student residence visa, the medical fitness test, or the Emirates ID. The entry permit procedure, bank accounts, and lease registration on TAMM are not covered, because no official page that could be opened supports them. The dates on the plan are the team's suggested order and have no official basis, and the interface says so on every card.

The benchmarking rests on page text fetched on 2 October 2026. The TAMM website could not be opened, and it is unconfirmed whether TAMM or UAE Pass serves a student before the Emirates ID is issued. The stated white space is absence of evidence. No OpenAI page documents text quality per language. The prototype offers English, Spanish, Portuguese, Chinese, and Japanese; the team tested English end to end and Spanish in one probe, and it has not reviewed the other languages with a speaker.

10. Adoption path and next steps

The university is the natural adopter, because it sponsors the student and issues the study certificate [2]. The team lead works in ADU's International Students Recruitment Office, which gives the team access to that stakeholder. The next step is a pilot with one university international office for one intake, in which the office owns and reviews the knowledge base and the team measures how many students complete the first-week steps on time. That pilot is proposed and not agreed.

Three technical steps follow the pilot. First, each university adds its own entries and current fees, which replaces the 2018 form. Second, the deployment moves to OpenAI's UAE data residency option [28]. Third, buddy matching is built on a university's student clubs in place

of the sample data in the prototype. The request to the Hub71 community is an introduction to the TAMM team or to a second Abu Dhabi university, so that Dalil's handoff into government services can be confirmed with the service owner.

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